Rajkumar Pawar

+91-9146530955 | pawarrajkumar0707@gmail.com | linkedin.com/in/rajkumarpawar07 | github.com/rajkumarpawar07

WORK EXPERIENCE

Software Engineer – Accenture September 2024 - Present

Pune, Maharashtra, India

Project: Agentic Documentation Assistant *December 2025 - Present*

- Architected a production **Agentic AI system** using Google Agent Development Kit (ADK) and Model Context Protocols (MCPs) to retrieve precise answers from internal Confluence documentation, cutting manual lookup time by **70%**.
- Built reusable **Terraform IaC modules** and **Azure DevOps CI/CD pipelines** for GCP platform, standardising provisioning across **10+ services**, migrating auth to **Workload Identity Federation** to eliminate credential leakage, and reducing manual config errors by **80%**.
- Engineered security data pipelines with **Azure Data Factory** in production; automated scheduling via Apache Airflow achieving **99.5%+** pipeline uptime.

Project: RAG-powered aircraft maintenance tool (Charlie) *September 2024 - December 2025*

- Designed and deployed a production **RAG-powered aircraft maintenance tool (Charlie)** on **Google Cloud (Vertex AI, Gemini, BigQuery)** for Europe's one of the biggest airline company's **15,000+ engineers**, enabling technicians to search part numbers, diagnose damage, and retrieve repair procedures across **550+ aircraft** from Airbus, Boeing, GE, and CFM, reducing lookup time by **∼60%**.
- Architected a **hybrid retrieval engine** combining **Vertex AI Vector Search (semantic)**, **BM25 keyword matching (exact P/N)**, and **BigQuery structured lookups** with **Reciprocal Rank Fusion** and **cross-encoder re-ranking**, safety-critical responses with full source traceability meeting EASA Part 145 and EU AI Act compliance.
- Built an **end-to-end document ingestion pipeline** integrating **OpenText Documentum** with **event-driven sync**, **ATA chapter-aware intelligent chunking**, and **attribute-based access control (ABAC)** enforcing **document-level ACLs** across retrieval and generation, ensuring only approved, revision-controlled documentation surfaces to authorized roles.
- Implemented a **multi-layer grounding verification framework** with automated **quality gates** (**near-zero hallucination rate**, **>99.5% part number accuracy**, **>95% faithfulness**) within the **Google GenAI Factory**, accelerating **development-to-production cycles** by **35%** through reusable RAG templates and **CI/CD-integrated evaluation pipelines**.

Software Engineer Intern – Argan Technology Services Pvt. Ltd. November 2023 - July 2024

Pune, Maharashtra, India

- Engineered AEGIS, a full-stack HR management system **(Flutter, FastAPI, MongoDB)** for **50+ employees** across 3 departments, reducing administrative tasks by **30%** and publishing the app to the Google Play Store.
- Built major features like employee live location tracking, payroll management, time and attendance, leave management and automated backend deployment **(GitHub Actions CI/CD)** for FastAPI services on **AWS EC2.**

PROJECTS

Playflex: AI-Powered Video Streaming Platform - GitHub

- Designed an end-to-end ML pipeline involving data cleaning, feature engineering, TF-IDF vectorization, model training, hyperparameter tuning, and evaluation on **100,000+** IMDb reviews, achieving **85%** sentiment classification accuracy using **Scikit-learn.**
- Built FastAPI for RESTful APIs to serve sentiment predictions and app data, reducing API response time by **40%**, within an Alan AI voice-assistant-integrated Flutter and Firebase app.

Advanced RAG Architectures Suite - GitHub

- Architecting a comprehensive suite of **7 production-grade RAG pipelines** from scratch, progressing from baseline vector search to fully autonomous multi-agent reasoning and knowledge graph traversal.
- Engineered an Agentic RAG system using **LangGraph** to orchestrate a ReAct state machine, routing searches between Qdrant and real-time web search (Tavily); implemented Corrective RAG (CRAG) & Self-RAG with HyDE to reduce hallucinations.

Autonomous Multi-Agent Content Generator *(Python, LangGraph, OpenAI, Gemini, Streamlit)*

- Architected a multi-agent AI workflow using **LangGraph** to autonomously plan, research, and write comprehensive markdown articles, replacing zero-shot prompting with an orchestrator-worker pattern.
- Engineered a dynamic routing engine utilizing the Tavily API to evaluate if topics required real-time context, fetching and de-duplicating web sources to heavily reduce LLM hallucinations.

TECHNICAL SKILLS

- **Generative AI:** RAG, LangChain, LangGraph, Google ADK, MCP, Agentic AI, LLM Fine-tuning
- **Machine Learning:** Scikit-learn, XGBoost, TensorFlow, PyTorch, NLP
- **Cloud Platforms:** GCP (Vertex AI, GKE, BigQuery, Cloud Functions, Pub/Sub, Cloud Run), AWS (Bedrock, Lambda, Kendra, Lex, S3), Azure (Data Factory, DevOps)
- **Programming Languages:** Python, Golang, SQL, Bash/YAML
- **MLOps & DevOps:** Docker, Kubernetes, Terraform, GitHub Actions, Azure DevOps, Jenkins, Git
- **Databases & Vector Stores:** PostgreSQL, MongoDB, Redis, FAISS, Qdrant, pgvector
- **Monitoring & Observability:** Prometheus, Grafana, LangSmith, Ragas, OpenTelemetry

EDUCATION

JSPM's Rajarshi Shahu College of Engineering, Pune 2020 - 2024

B.Tech, Electronics & Telecommunication Engineering CGPA: 9.07 / 10

CERTIFICATIONS

GCP : Associate Cloud Engineer | Professional Cloud Architect | Professional ML Engineer | Professional Data Engineer

Azure : Administrator Associate (AZ-104) | DevOps Engineer Expert (AZ-400)